Corrected 10-Slide Presentation Script

Based on your actual 10-slide presentation • exactly 4 speaking lines per slide

Slide 1 — Title and Team

Presenter: **B. Suhas**

1. Good morning everyone.
2. Our project is titled “Mathematical Modelling, Verification, and Analysis of a Secure Public-Key Messaging Framework.”
3. Our team members are B. Suhas, Samiksh, Vardhan, and Akhilesh.
4. Today, we will explain how mathematics helps us model and analyse secure messaging.

Slide 2 — Introduction to Secure Messaging

Presenter: **B. Suhas**

1. Secure messaging helps protect information sent between people.
2. Confidentiality keeps a message private, and integrity helps protect it from unwanted changes.
3. Authentication helps confirm the identity of the communicating party.
4. Good key management is also important for secure communication.

Slide 3 — Mathematical Model Using Sets and Functions

Presenter: **B. Suhas**

1. We represent the original plaintext as P and the encrypted ciphertext as C.
2. Encryption is written as C = E(P, Kpub), using the receiver’s public key.
3. Decryption is written as P′ = D(C, Kpriv), using the matching private key.
4. Correctness means the decrypted message P′ should be equal to the original message P.

Slide 4 — Key-Space and Counting

Presenter: **Samiksh**

1. A key-space means all the possible codes or keys that can be chosen.
2. A four-digit decimal code has 10,000 possible combinations.
3. An eight-digit decimal code has 100 million possible combinations.
4. This demonstrates counting, but the number of combinations alone does not guarantee security.

Slide 5 — RSA Key Generation

Presenter: **Samiksh**

1. RSA is a public-key cryptography method based on number theory.
2. For our small example, we choose the prime numbers 5 and 11.
3. Multiplying them gives n = 55, and the value of phi of n is 40.
4. The public key is (3, 55), and the private key is (27, 55); these tiny values are only for learning.

Slide 6 — RSA Encryption and Decryption

Presenter: **Samiksh**

1. Let the original message number be m = 7.
2. Encrypting gives $c = 7^3 \bmod 55 = 13$.
3. Decrypting ciphertext 13 with the private key gives back 7.
4. This toy example shows the process, but it is not secure enough for real communication.

Slide 7 — Graph Theory Model

Presenter: **Vardhan**

1. Graph theory can represent users and communication links in a network.
2. Each user is represented by a vertex, and each link is represented by an edge.
3. A path in the graph shows one possible route for a message between users.
4. Connectivity tells us whether users can reach one another, but it does not prove that the message is secure.

Slide 8 — Boolean Logic and Verification

Presenter: **Vardhan**

1. Boolean logic helps us express simple verification conditions using true or false values.
2. Let A mean the key pair is valid, B mean the ciphertext is valid, and C mean verification succeeds.
3. We write the combined condition as S = A AND B AND C.
4. In this simplified model, S is true only when all three conditions are true.

Slide 9 — Security Analysis

Presenter: **Vardhan**

1. A mathematically correct system is not automatically secure in real-world use.
2. A secure messaging system needs confidentiality, integrity, and authentication.
3. It also needs careful key management and secure key distribution.
4. Modern systems use properly implemented cryptographic techniques, such as authenticated encryption and digital signatures.

Slide 10 — Performance Analysis and Conclusion

Presenter: **Akhilesh**

1. We can analyse performance by considering message size, key size, and communication overhead.
2. We can also study the computation needed for encryption and decryption.
3. Our project uses mathematical models to discuss correctness, security requirements, and performance trade-offs without inventing benchmark results.
4. To conclude, mathematics helps us understand secure messaging. Thank you for listening.